

Ambali Alade Odebowale

PhD, Electrical Engineering · Nanophotonics and Radiative Heat Transfer

Canberra, ACT 2913, Australia · +61 421 605 911 · aodebowale.c@gmail.com

[Google Scholar](#) · ORCID [0000-0002-6645-7666](#) · [linkedin.com/in/odebowale-ambali-phd](#) · [github.com/Alademighty](#)

RESEARCH PROFILE

Electrical engineer and applied physicist working at the boundary of nanophotonics and thermal physics. My research uses **machine learning to invert the design problem** in near-field radiative heat transfer, where the parameter spaces governing thermophotovoltaic emitters are too large for conventional sweeps. Eleven peer-reviewed publications, five as first author, including *Physical Review Applied* and *Advanced Functional Materials*. PhD recognised by the UNSW Dean’s Award for Outstanding Theses. Eight years of teaching across four institutions in three countries. Seeking a postdoctoral appointment or lectureship.

EDUCATION

PhD, Electrical Engineering May 2021 – April 2025

The University of New South Wales, Canberra (ADFA), Australia

Thesis: *[insert thesis title]*

Supervisors: Prof. Andrey E. Miroshnichenko and Prof. Haroldo T. Hattori

Dean’s Award for Outstanding PhD Theses, 2025

MSc, Electrical Engineering (Electronics major) Sept 2016 – Feb 2020

King Saud University, Riyadh, Saudi Arabia

BTech, Electronic and Electrical Engineering Aug 2008 – April 2014

Ladoke Akintola University of Technology, Ogbomosho, Nigeria

Summer School, Printed Hybrid Photonic and Electronic Structures July – Aug 2022

Karlsruhe School of Optics and Photonics, Germany

AWARDS AND HONOURS

Dean’s Award for Outstanding PhD Theses 2025

The University of New South Wales

Best Higher Degree Research Student, School of Engineering and Technology 2024

UNSW Canberra

University International Postgraduate Award (UIPA) — full tuition and stipend 2021 – 2024

UNSW Canberra

Travel Award 2022

Karlsruhe School of Optics and Photonics, Germany

Government of Saudi Arabia Scholarship — full funding 2016 – 2019

King Saud University

PUBLICATIONS

Eleven peer-reviewed journal articles and reviews; five as first author (shown in bold). Citation metrics and any newer work are available on [Google Scholar](#).

1. **Odebowale, A. A.**; Berhe, A.; Ogundare, R. T.; Abdo, S.; Abdulghani, A.; Hattori, H. T.; Miroshnichenko, A. E. “Advances in radiative heat transfer: bridging far-field fundamentals and emerging near-field innovations.” *Advanced Functional Materials*, 35(27), 2421051 (2025). doi:10.1002/adfm.202421051.
2. **Odebowale, A. A.**; Berhe, A. M.; Somaweera, D.; Wang, H.; Lei, W.; Miroshnichenko, A. E.; Hattori, H. T. “Advances in 2D photodetectors: materials, mechanisms and applications.” *Micromachines*, 16(7), 776 (2025). doi:10.3390/mi16070776.

3. Somaweera, D.; Abdulghani, A.; **Odebowale, A. A.**; Berhe, A. M.; Weerasinghe, M. I. U.; As'ham, K.; Al Ani, I. A. M.; Dumlao, M. C.; Miroshnichenko, A. E.; Hattori, H. T. "Rydberg atom-based sensors: principles, recent advances and applications." *Photonics*, 12(12), 1228 (2025). doi:10.3390/photonics12121228.
4. Abdulghani, A.; Abdo, S.; As'ham, K.; **Odebowale, A. A.**; Miroshnichenko, A. E.; Hattori, H. T. "Directional coupling of surface plasmon polaritons at exceptional points in the visible spectrum." *Materials*, 18(24), 5595 (2025).
5. Abdo, S.; As'ham, K.; **Odebowale, A. A.**; Akter, S.; Abdulghani, A.; Al Ani, I. A. M.; Hattori, H. T.; Miroshnichenko, A. E. "A near-ultraviolet photodetector based on the TaC:Cu / 4H silicon carbide heterostructure." *Applied Sciences*, 15(2), 970 (2025). doi:10.3390/app15020970.
6. Berhe, A. M.; Shvedov, V.; **Odebowale, A. A.**; Hattori, H. T.; Miroshnichenko, A. E. "All-pass Si₃N₄ metasurface filter for advanced photonic applications: metalenses, vortex beams and holography." *Journal of Optics*, 28(1), 015102 (2025).
7. Berhe, A. M.; **Odebowale, A. A.**; Koshelev, K.; Hattori, H. T.; Miroshnichenko, A. E. "Characterising nanoscale spatiotemporal defects of multi-layered MoSe₂ in hyper-temporal transient nanoscopy." *Nanophotonics*, (2025). doi:10.1515/nanoph-2025-0163.
8. **Odebowale, A. A.**; As'ham, K.; Hattori, H. T.; Miroshnichenko, A. E. "Design and optimization of near-field thermophotovoltaic systems using deep learning." *Physical Review Applied*, 21, 064031 (2024). doi:10.1103/PhysRevApplied.21.064031.
9. **Odebowale, A. A.**; As'ham, K.; Berhe, A. M.; Alim, N.; Hattori, H. T.; Miroshnichenko, A. E. "Near-field radiative thermal rectification assisted by a Bi₂Se₃ sheet." *International Communications in Heat and Mass Transfer*, 157, 107707 (2024). doi:10.1016/j.icheatmasstransfer.2024.107707.
10. **Odebowale, A. A.**; Berhe, A. M.; Hattori, H. T.; Miroshnichenko, A. E. "Modeling and analysis of a radiative thermal memristor." *Applied Sciences*, 14(6), 2633 (2024). doi:10.3390/app14062633.
11. **Odebowale, A. A.**; Abdulghani, A.; Berhe, A. M.; Somaweera, D.; Akter, S.; Abdo, S.; As'ham, K.; Saadabad, R. M.; Tran, T. T.; Bishop, D. P.; Solntsev, A. S.; Miroshnichenko, A. E.; Hattori, H. T. "Emerging low detection limit of optically activated gas sensors based on 2D and hybrid nanostructures." *Nanomaterials*, 14(18), 1521 (2024). doi:10.3390/nano14181521.

RESEARCH EXPERIENCE

Doctoral Researcher

2021 – 2025

School of Engineering and Technology, UNSW Canberra

- Built a deep-learning framework — fully connected and convolutional networks — to predict spectral heat flux and optimise near-field thermophotovoltaic emitters, replacing exhaustive fluctuational-electrodynamics parameter sweeps. Published in *Physical Review Applied*.
- Proposed and modelled a **radiative thermal memristor** with history-dependent conductance, with applications in contactless thermal sensing and neuromorphic thermal computing.
- Demonstrated near-field radiative thermal rectification assisted by a Bi₂Se₃ sheet, contributing a route to non-reciprocal thermal components.
- Led a field-wide review of radiative heat transfer from far-field fundamentals to near-field device innovation, published in *Advanced Functional Materials*.
- Collaborated across the photonics group on photodetector, gas-sensing and metasurface work, co-authoring six further papers.

Researcher A

Nov 2017 – Dec 2021

Prince Sultan Advanced Technology Research Institute (PSATRI), King Saud University, Riyadh

- Designed, fabricated and tested infrared and electro-optical sensors, focal plane arrays and camera systems in the Electro-Optic Laboratory.
- Conducted and published peer-reviewed research within multidisciplinary project teams.
- Mentored undergraduate and postgraduate students on research projects, and presented findings at conferences and workshops.

TEACHING EXPERIENCE

Lecturer (casual)

Aug – Nov 2024

Wentworth Institute of Higher Education, Canberra, Australia

- Designed and delivered the Programming Fundamentals curriculum, teaching Python from first principles to students with no prior coding experience.
- Lectured and tutored across information systems and IT fundamentals; set, marked and moderated all assessment.
- Maintained cohort progress records and reported on learning outcomes.

Lecturer (undergraduate and postgraduate)

2017 – 2021

King Saud University, Riyadh, Saudi Arabia

- Taught Microelectronics and Very Large Scale Integrated (VLSI) Circuit Design to undergraduate and graduate cohorts, alongside a full research load.

Technical Instructor (part-time)

Jan – Dec 2021

Saudi Electric Services Polytechnic, Riyadh, Saudi Arabia

- Delivered vocational courses in Electrical Distribution and Maintenance, Overhead Line, and Solar PV installation and design, combining lectures with supervised practical training and field assessment.

Mathematics and Physics Tutor

Aug 2022 – Sept 2024

Superprof and EzyMath, Canberra, Australia

- One-to-one tutoring in calculus, linear algebra, mechanics and electromagnetism for secondary and university students, with custom problem sets and progress tracking.

Laboratory Assistant

July – Dec 2022

UNSW Canberra

- Prepared and ran undergraduate laboratory sessions, guided students through experimental procedure and instrumentation, and enforced laboratory safety protocol.

PROFESSIONAL SERVICE

Peer reviewer

current

IEEE Journal of Quantum Electronics · Silicon

Vice President, Optica Student Chapter

2023 – 2025

UNSW Canberra (ADFA)

- Organised seminars and outreach activities for the chapter's photonics research community.

TECHNICAL SKILLS

Computational methods Fluctuational electrodynamics · near-field radiative transfer · FDTD and RCWA · transfer-matrix method · semiconductor device modelling · inverse design

Machine learning Fully connected and convolutional networks · surrogate modelling for photonic design · data manipulation and visualisation

Programming and software Python (NumPy, SciPy) · MATLAB · Lumerical scripting · Mathematica · Cadence · PSpice · L-Edit · AutoCAD · SQL · LaTeX

Experimental Infrared and electro-optical sensor characterisation · focal plane arrays · photodetector testing · electrical safety testing and fault diagnosis

PROFESSIONAL MEMBERSHIP, CERTIFICATION AND TRAINING

Chartered Professional Engineer, Engineers Australia (EA ID 8913867, ANZSCO 233411)

2024

Member, Optica (formerly OSA)

since 2021

Member, IEEE

Machine Learning using Python; Advanced Python for data manipulation and visualisation — Intersect Australia

NSW Electrical Test and Tag certification (ID 15047566)

Engineering for New Energy (APA Group) and Data Analytics (Deloitte Australia) job simulations — Forage

since 2013

2024

2024

2025

ADDITIONAL PROFESSIONAL EXPERIENCE

Industry and technical roles held before and alongside the doctorate.

- Technical Officer (casual), UNSW Canberra

July 2024 – present

Electrical safety testing and tagging across campus, fault diagnosis on electrical equipment, and risk assessment to workplace safety standards.
- Service Electrical Engineer, Adekass Technical Services Ltd, Lagos, Nigeria

Feb 2015 – Aug 2016

Maintenance, installation and fault diagnosis on industrial diesel generators (Perkins, Cummins, Caterpillar) and electronic modular control panels.
- Solar PV Designer and Installer, ANSA Power Systems Ltd, Ibadan, Nigeria

March 2013 – Jan 2015

Site assessment, system sizing, component selection and end-to-end project management for residential and industrial solar PV installations.

REFEREES

Confirm each referee's consent and contact details before submitting.

- Prof. Andrey E. Miroshnichenko — Doctoral supervisor

School of Engineering and Technology, UNSW Canberra · [email] · [phone]
- Prof. Haroldo T. Hattori — Doctoral co-supervisor

School of Engineering and Technology, UNSW Canberra · [email] · [phone]
- [Third referee] — [Relationship]

[Institution] · [email] · [phone]